

Chapter 3 Conquering the ECG

Besides the stethoscope, the electrocardiogram (ECG) is the oldest and most enduring tool of the cardiologist. A basic knowledge of the ECG will enhance the understanding of **cardiology** (not to mention this book).

Electrocardiography

At every beat, the heart is depolarized to trigger its contraction. This electrical activity is transmitted throughout the body and can be picked up on the skin. This is the principle behind the ECG. An ECG machine records this activity via electrodes on the skin and displays it graphically. An ECG involves attaching 10 electrical cables to the body: one to each limb and six across the chest.

ECG terminology has two meanings for the word "lead":

- the cable used to connect an electrode to the ECG recorder
- the electrical view of the heart obtained from any one combination of electrodes

Carrying out an ECG

1. Ask the patient to undress down to the waist and lie down
2. Remove excess hair where necessary
3. Attach limb leads (anywhere on the limb)
4. Attach the chest leads (see [Figure 1](#)) as follows:
 - V1 and V2: either side of the sternum on the fourth rib (count down from the sternal angle, the second rib insertion)
 - V4: on the apex of the heart (feel for it)
 - V3: halfway between V2 and V4
 - V5 and V6: horizontally laterally from V4 (not up towards the axilla)
5. Ask the patient to relax
6. Press record

The standard ECG uses 10 cables to obtain 12 electrical views of the heart. The different views reflect the angles at which electrodes "look" at the heart and the direction of the heart's electrical depolarization.

Limb leads

Three bipolar leads and three unipolar leads are obtained from three electrodes attached to the left arm, the right arm, and the left leg, respectively. (An electrode is also attached to the right leg, but this is an earth electrode.) The bipolar limb leads reflect the potential difference between two of the three limb electrodes:

- **lead I**: right arm–left arm
- **lead II**: right arm–left leg
- **lead III**: left leg–left arm

The unipolar leads reflect the potential difference between one of the three limb electrodes and an estimate of zero potential – derived from the remaining two limb electrodes. These leads are known as augmented leads. The augmented leads and their respective limb electrodes are:

- **aVR lead:** right arm
- **aVL lead:** left arm
- **aVF lead:** left leg

Chest leads

Another six electrodes, placed in standard positions on the chest wall, give rise to a further six unipolar leads – the chest leads (also known as precordial leads), V1–V6. The potential difference of a chest lead is recorded between the relevant chest electrode and an estimate of zero potential – derived from the average potential recorded from the three limb leads.

Planes of view

The limb leads look at the heart in a vertical plane (see [Figure 2](#)), whereas the chest leads look at the heart in a horizontal plane. In this way, a three-dimensional electrical picture of the heart is built up (see [Table 1](#)).

Performing Dogs

British physiologist Augustus D Waller of St Mary's Medical School, London, published the first human electrocardiogram in the British Medical Journal in 1888. It was recorded from Thomas Goswell, a technician in the laboratory, using a capillary electrometer. After that, Waller used a more available subject for his demonstrations – his dog Jimmy, who would patiently stand with his paws in glass jars of saline.

Depolarization of the heart

The route that the depolarization wave takes across the heart is outlined in [Figure 3](#). The sinoatrial node (SAN) is the heart's pacemaker. From the SAN, the wave of depolarization spreads across the atria to the atrioventricular node (AVN). The impulse is delayed briefly at the AVN and atrial contraction is completed.

The wave of depolarization then proceeds rapidly to the bundle of His where it splits into two pathways and travels along the right and left bundle branches. The impulse travels the length of the bundles along the interventricular septum to the base of the heart, where the bundles divide into the Purkinje system. From here, the wave of depolarization is distributed to the ventricular walls and initiates ventricular contraction.

The ECG trace

The ECG machine processes the signals picked up from the skin by electrodes and produces a graphic representation of the electrical activity of the patient's heart. The basic pattern of the ECG is logical:

- electrical activity towards a lead causes an upward deflection
- electrical activity away from a lead causes a downward deflection
- depolarization and repolarization deflections occur in opposite directions

The basic pattern of this electrical activity was first discovered over a hundred years ago. It comprises three waves, which have been named P, QRS (a wave complex), and T (see [Figure 4](#)).

P wave

The P wave is a small deflection wave that represents atrial depolarization.

PR interval

The PR interval is the time between the first deflection of the P wave and the first deflection of the QRS complex.

QRS wave complex

The three waves of the QRS complex represent ventricular depolarization. For the inexperienced, one of the most confusing aspects of ECG reading is the labeling of these waves. The rule is: if the wave immediately after the P wave is an upward deflection, it is an R wave; if it is a downward deflection, it is a Q wave:

- small Q waves correspond to depolarization of the interventricular septum. Q waves can also relate to breathing and are generally small and thin. They can also signal an old myocardial infarction (in which case they are big and wide)
- the R wave reflects depolarization of the main mass of the ventricles –hence it is the largest wave
- the S wave signifies the final depolarization of the ventricles, at the base of the heart

ST segment

The ST segment, which is also known as the ST interval, is the time between the end of the QRS complex and the start of the T wave. It reflects the period of zero potential between ventricular depolarization and repolarization.

T wave

T waves represent ventricular repolarization (atrial repolarization is obscured by the large QRS complex).

Wave direction and size

Since the direction of a deflection, upward or downward, is dependent on whether the electrical activity is going towards or away from a lead, it differs according to the orientation of the lead with respect to the heart (see [Figure 5](#)).

The ECG trace reflects the net electrical activity at a given moment. Consequently, activity in one direction is masked if there is more activity, eg, by a larger mass, in the other direction. For example, the left ventricle muscle mass is much greater than the right, and therefore its depolarization accounts for the direction of the biggest wave.

Interpreting the ECG

A normal ECG tracing is provided in [Figure 6](#). The only way to become confident at reading ECGs is to practice. It is important to be methodical – every ECG reading should start with an assessment of the rate, rhythm, and axis. This approach always reveals something about an ECG, regardless of how unusual it is.

Rate

Identify the QRS complex (this is generally the biggest wave); count the number of large squares between one QRS wave and the next; divide 300 by this number to determine the rate (see [Table 2](#)).

Rhythm

P waves are the key to determining whether a patient is in sinus rhythm or not. If P waves are not clearly visible in the chest leads, look for them in the other leads. The presence of P waves immediately before every QRS complex indicates sinus rhythm. If there are no P waves, note whether the QRS complexes are wide or narrow, regular or irregular.

No P waves and irregular narrow QRS complexes

This is the hallmark of atrial fibrillation (see [Figure 7](#)). Sometimes the baseline appears "noisy" and sometimes it appears entirely flat. However, if there are no P waves and the QRS complexes appear at randomly irregular intervals, the diagnosis is atrial fibrillation.

Sawtooth P waves

A sawtooth waveform signifies atrial flutter (see [Figure 8](#)). The number of atrial contractions to one ventricular contraction should be specified.

Axis

The axis is the net direction of electrical activity during depolarization. It is altered by left ventricular or right ventricular hypertrophy or by bundle branch blocks. It is a very straightforward measurement that, once it has been grasped, can be calculated instantaneously:

- find the QRS complex in the I and aVF leads (because these look at the heart at 0° and $+90^\circ$, respectively)
- determine the net positivity of the QRS wave from each of the two leads by subtracting the S wave height (the number of small squares that it crosses as it dips below the baseline – if it does) from the R wave height (the number of small squares that it crosses as it rises) (see [Figure 9a](#) and [9b](#))
- plot out the net sizes of these QRS waves against each other on a vector diagram (see [Figure 9c](#)). For the I lead, plot net positives to the right and net negatives to the left; for the aVF lead, plot positive downwards and negative upwards
- the direction of the endpoint from the starting point represents the axis or predominant direction of electrical depolarization (determined primarily by the muscle mass of the left ventricle). It is expressed as an angle and can be estimated quite easily (normal is 0° – 120°)

Human Resuscitation

The first electrical resuscitation of a human took place (almost certainly) in 1872. The resuscitation of a drowned girl with electricity is described by Guillaume Benjamin Amand Duchenne de Boulogne, a pioneering neurophysiologist, in the third edition of his textbook on the medical uses of electricity. Although it is sometimes described as the first artificial pacing, the stimulation was of the phrenic nerve and not the myocardium.

ECG abnormalities

This section discusses the most important and most frequently encountered ECG abnormalities.

Normal variations

- Small Q waves and inverted T waves in lead III often disappear on deep inspiration. Occasional septal Q waves can be seen in other leads.
- ST elevation following an S wave ("high take off") is common in leads V2–V4 and is quite normal. Differentiating this from pathological ST elevation can be difficult and relies on the patient's history and the availability of a previous ECG. These "repolarization abnormalities" are more common in the young and in athletes.
- T-wave inversion is common in Afro-Caribbean blacks.
- U waves – small extra waves following T waves – are seen in hypokalemic patients, but can also represent a normal variant.
- Ventricular extrasystoles – no P waves, broad and abnormal QRS complexes, and T waves interspersed between normal sinus rhythm – sometimes occur and do not require further investigation unless they are associated with symptoms (such as dizziness, palpitations, exercise intolerance, chest pain, shortness of breath) or occur several times every minute.

Pathological variations

Long PR interval

A distance of more than five small squares from the start of the P wave to the start of the R wave (or Q wave if there is one) constitutes first-degree heart block (see Figure 10). It rarely requires action, but in the presence of other abnormalities might be a sign of hyperkalemia, digoxin toxicity, or cardiomyopathy.

EKG or ECG?

There is some debate over exactly who invented the electrocardiogram. The Dutch "K" (elektrokardiogram) is often used as a tribute to the Indonesian-born physician Wilhelm Einthoven who, while working in The Netherlands in 1924, received the Nobel prize for "the discovery of the mechanism of the electrocardiogram". In fact, it was Augustus Désiré Waller, a physician trained in Edinburgh, who presented – to the students of St Mary's Hospital medical school, London, at the introductory lecture of the 1888 academic year – his "cardiograph", the first ever ECG recording in man. It was some years later, in 1901, that Wilhelm Einthoven reported his string galvanometer – with the limb leads labeled I, II, and III and the waves labeled P, QRS, and T as we know them today. In fact, although often credited with inventing the term electrocardiogram (which is why it is sometimes spelt the Dutch way), Einthoven credits Waller with this distinction in his 1895 publication in Pflügers Archives "Über die Form des menschlichen Elektrokardiogramms".

Q waves

A normal ECG has only very small Q waves. A downward deflection immediately following a P wave that is wider than two small squares or greater in height than a third of the subsequent R wave is significant: such Q waves can represent previous infarction (see Figure 11, previous page).

Large QRS complexes

Left ventricular hypertrophy (LVH) is one of the easiest and most useful diagnoses to make (see Figure 12). The Sokolow–Lyon index is the most commonly calculated index of estimation. Does the sum of the S wave in lead V1 (SV1) and the R wave in V6 (RV6) add up to more than 3.5 mV, ie, 35 small or seven big squares? If so, the patient has LVH by voltage criterion. Right ventricular hypertrophy is indicated by a dominant R wave in V1 (ie, R wave bigger than following S wave; Sokolow–Lyon index: R in V1 + S in V5 or V6 $\geq$ 1.05 mV) and right axis deviation.

Broad QRS complexes and strange-looking ECGs

A wide QRS complex despite sinus rhythm is the hallmark of bundle branch block. Left bundle branch block (LBBB) can cause the ECG to look extremely abnormal (see Figure 13). When faced with such an ECG – after calculating rate, rhythm, and axis – check the width of the QRS complex. If it is more than three small squares wide, it is abnormal. Bundle branch block can then be diagnosed by pattern recognition of the QRS complexes in the V1 and V6 leads (see Figure 14). New LBBB can be diagnostic of myocardial infarction (MI).

ST segment changes

The ST segment extends from the end of the S wave to the start of the T wave. It should be flat or slightly upsloping and level with the baseline. Elevation of more than two small squares in the chest leads or one small square in the limb leads, combined with a characteristic history, indicates the possibility of MI (see Figure 15, previous page). ST depression is diagnostic of ischemia (see Figure 16). It is worth noting that although ST elevation can localize the lesion (eg, anterior MI, inferior MI), ST depression cannot. Concave upwards ST elevation in all 12 leads is diagnostic of pericarditis.

T waves

In a normal ECG, T waves are upright in every lead except aVR. T-wave inversion can represent current ischemia or previous infarction (see Figure 17). In combination with LVH and ST depression, it can represent "strain". This form of LVH carries a poor prognosis.

Long QT interval

The QT interval should be less than half of the R–R interval. Calculation of the corrected QT (QTc) is generally not necessary and usually will have been done by the ECG machine (but beware of blindly believing any automated diagnostic system). Conditions associated with a long QT interval are outlined in [Table 3](#) (see [Figure 18](#)).

Long QT syndrome may also be drug-induced (see [Table 4](#), p. 32). Once this occurs, the responsible drug needs to be discontinued.

Pattern combinations

Digoxin

A reverse tick ST depression is characteristic and does not indicate toxicity. Digoxin toxicity can result in dysrhythmia.

Pulmonary embolism

Sinus tachycardia is seen in many patients with pulmonary embolism. New right bundle branch block (RBBB) or right axis deviation with "strain" can also indicate PE. The classic S_IQ_{III}T_{III} is less common.

Hyperkalemia

The absolute potassium level is less important than its rate of rise. ECG changes reflecting a rapid rise demand immediate action (see [Figures 19–21](#)). The level of danger increases as the ECG changes progress. The sequence generally follows the order:

- tall, tented T waves (see [Figure 19](#))
- lengthening of the PR interval
- reduction in the P-wave height
- widening of the QRS complex (see [Figure 20](#))
- "sinus" wave QRS pattern (see [Figure 21](#))

A sinus-wave QRS should be treated immediately with calcium chloride, whilst hyperkalemia associated with lesser ECG changes can be treated with insulin/glucose infusion.

PQRST?

Nobody knows for sure why these letters became standard. Certainly, mathematicians used to start lettering systems from the middle of the alphabet to avoid confusion with the frequently used letters at the beginning. Einthoven used the letters O to X to mark the timeline on his ECG diagrams and, of course, P is the letter that follows O. If the image of the PQRST diagram was striking enough to be adopted by researchers as a true representation of the underlying form, it would have been logical to continue the same naming convention when the more advanced string galvanometer started creating ECGs a few years later.

Further reading

1. Ashley EA, Raxwal VK, Froelicher VF. The prevalence and prognostic significance of electrocardiographic abnormalities. *Curr Probl Cardiol.* 2000;25:1–72. [PubMed: 10705558]
2. Hampton JR. *ECG Made Easy*. London: Churchill Livingstone, 1997.
3. Rautaharju PM. A hundred years of progress in electrocardiography. 1: Early contributions from Waller to Wilson. *Can J Cardiol.* 1987;3:362–74. [PubMed: 3322535]

Figures

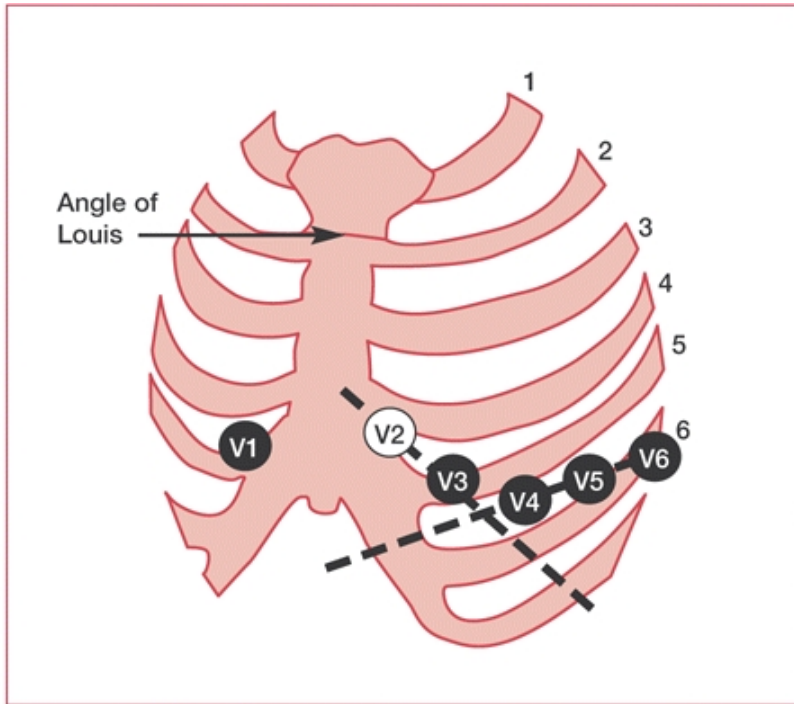


Figure 1

Standard attachment sites for chest leads.

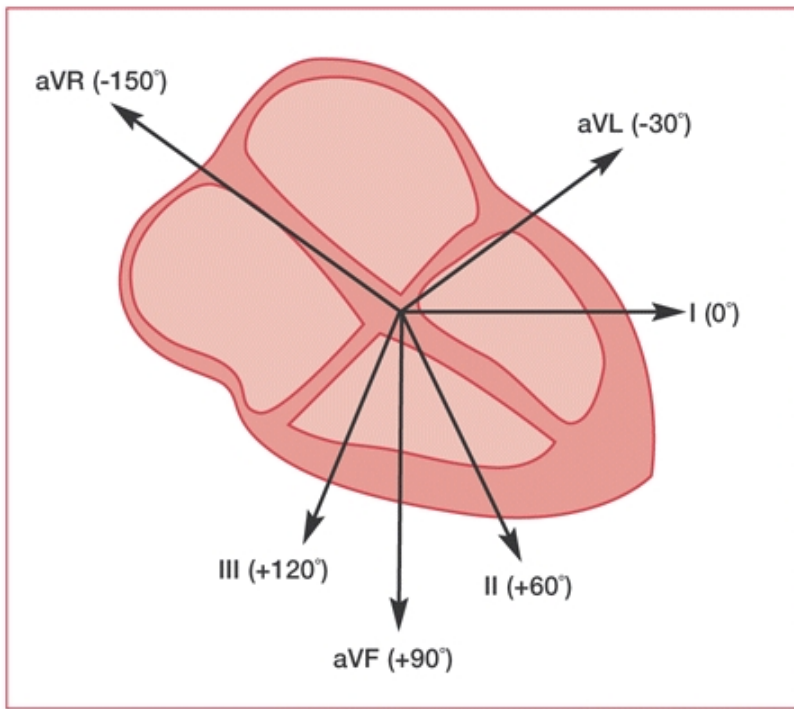


Figure 2

The limb leads looking at the heart in a vertical plane.

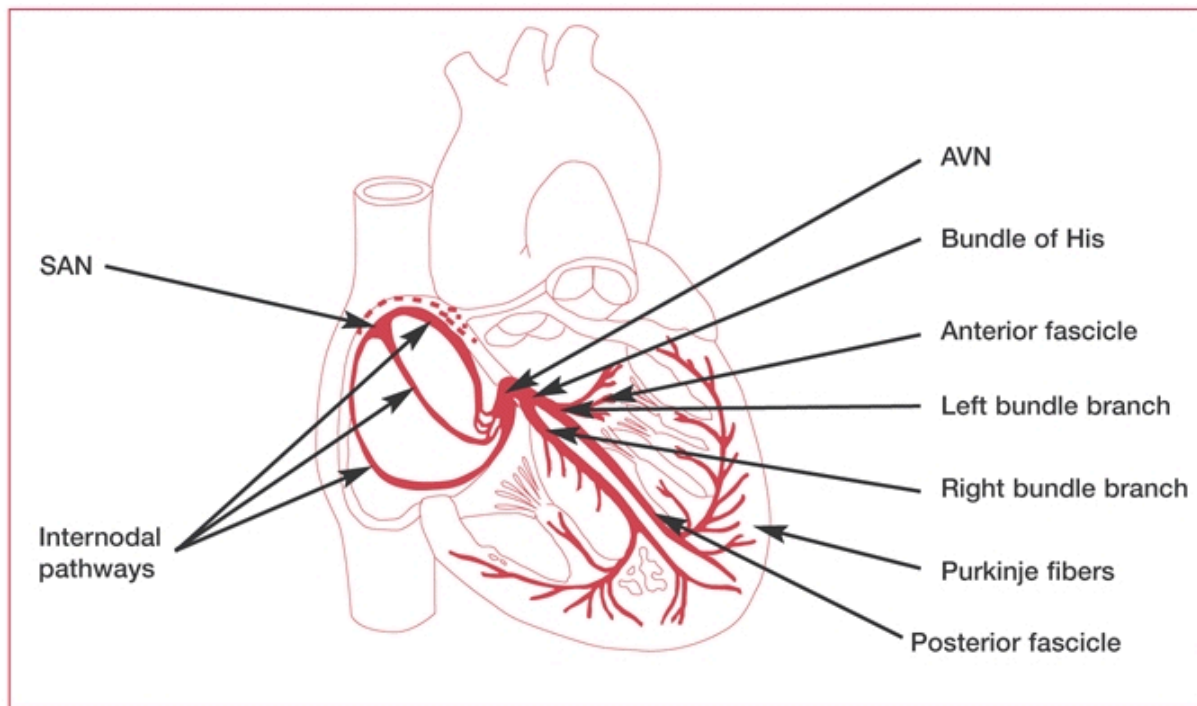


Figure 3

The cardiac depolarization route. AVN: atrioventricular node; SAN: sinoatrial node. Reproduced with permission from WB Saunders (Guyton A, Hall J. *Textbook of Medical Physiology*. Philadelphia: WB Saunders, 1996).

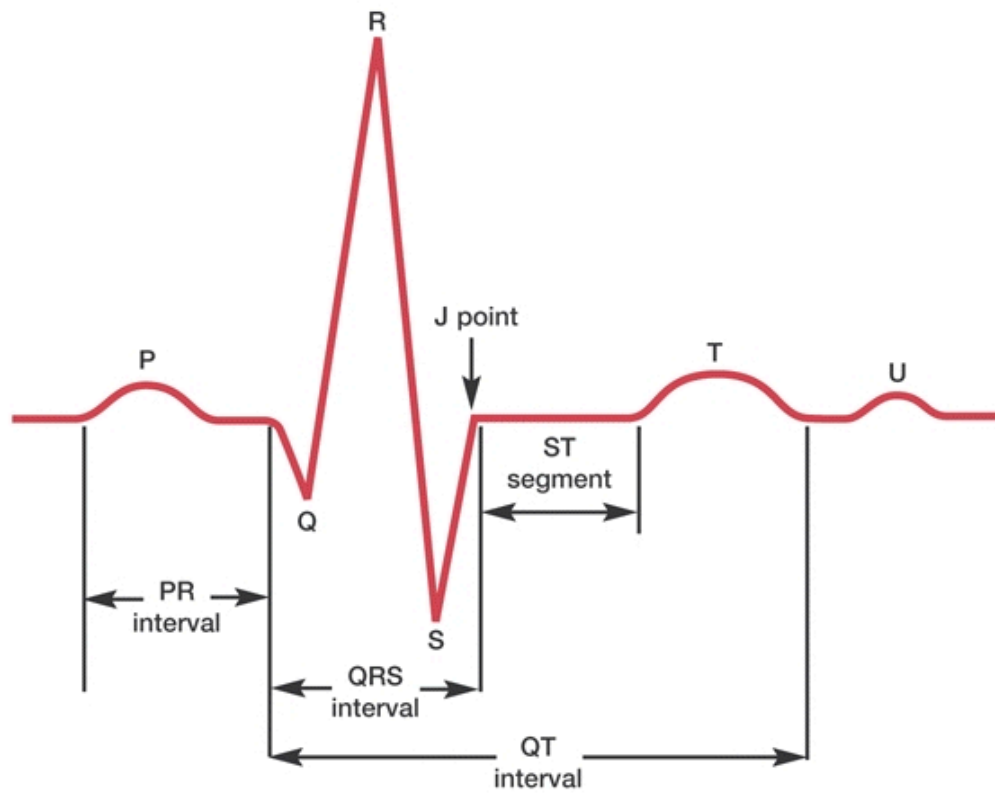


Figure 4

The basic pattern of electrical activity across the heart.

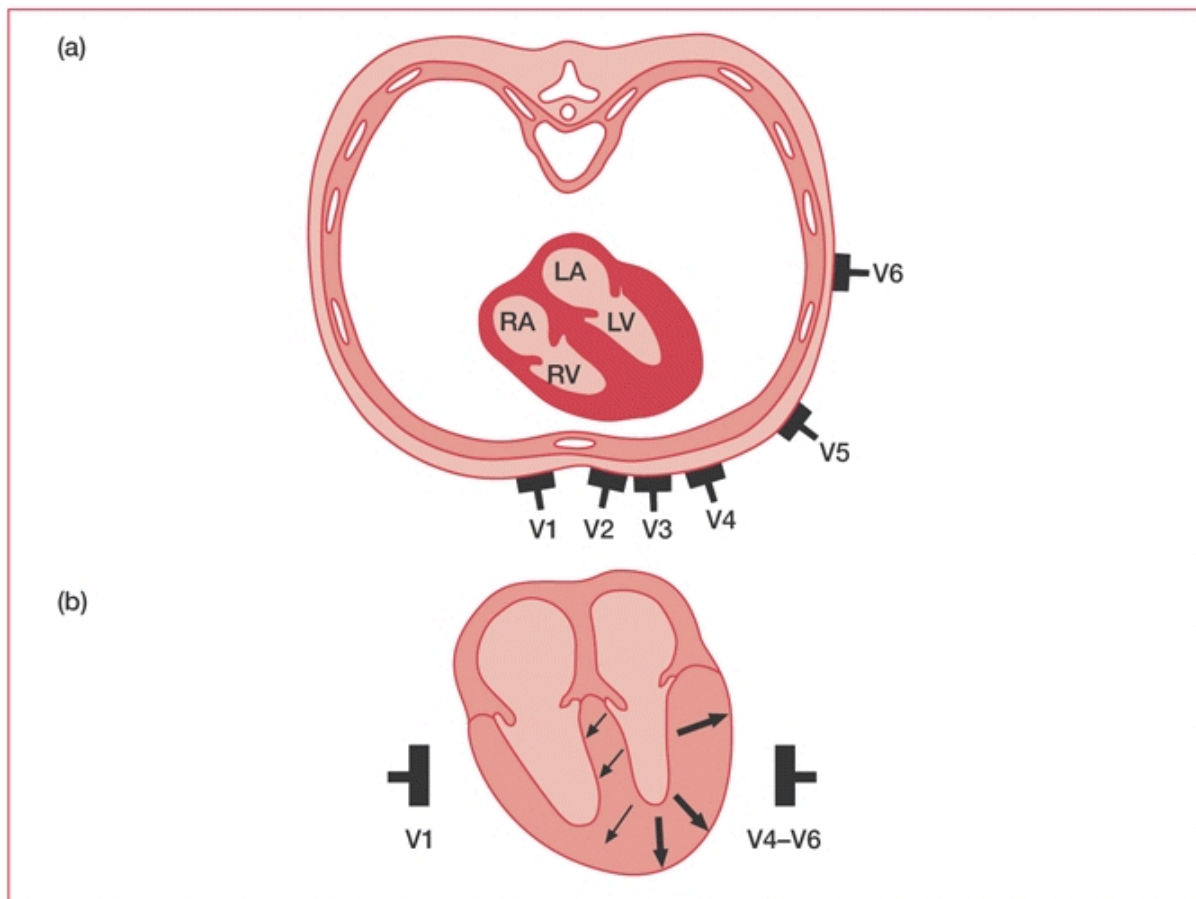


Figure 5

(a) A horizontal section through the chest showing the orientation of the chest leads with respect to the chambers of the heart. (b) In lead V1, depolarization of the interventricular septum occurs towards the lead, thus creating an upward deflection (R wave) on the ECG. It is followed by depolarization of the main mass of the LV, which occurs away from the lead, thus creating a downward deflection (S wave). This pattern is reversed for lead V6, explaining the different shapes of the QRS complex. This pattern should be checked in every ECG. LA: left atrium; LV: left ventricle; RA: right atrium; RV: right ventricle.

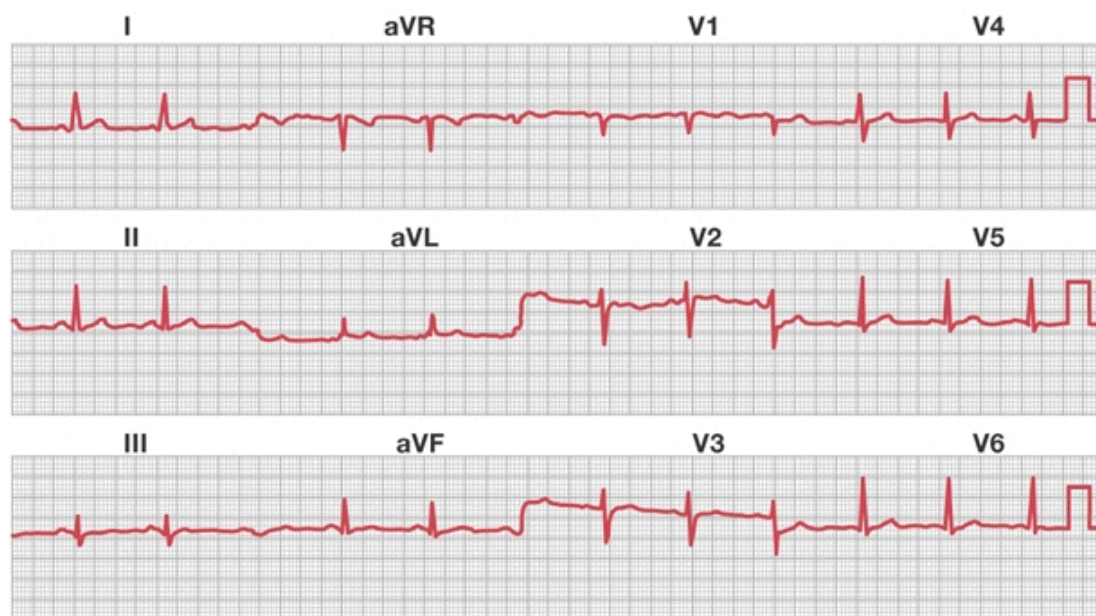


Figure 6

Example of a normal ECG.

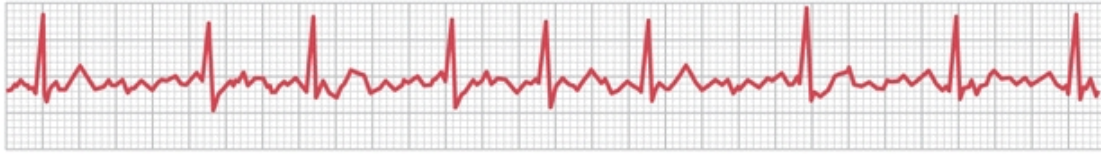


Figure 7

ECG demonstrating atrial fibrillation.

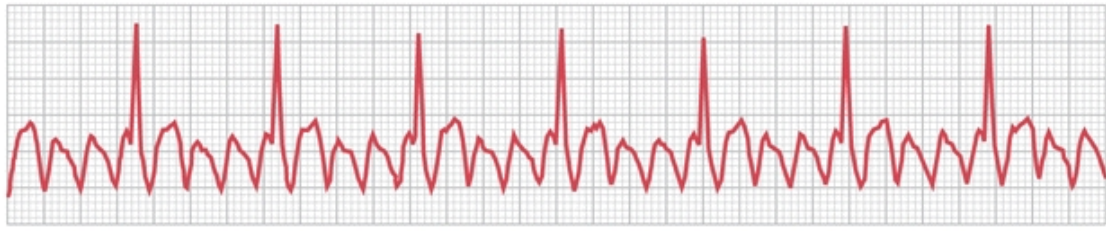


Figure 8

ECG demonstrating atrial flutter – note the characteristic sawtooth waveform.

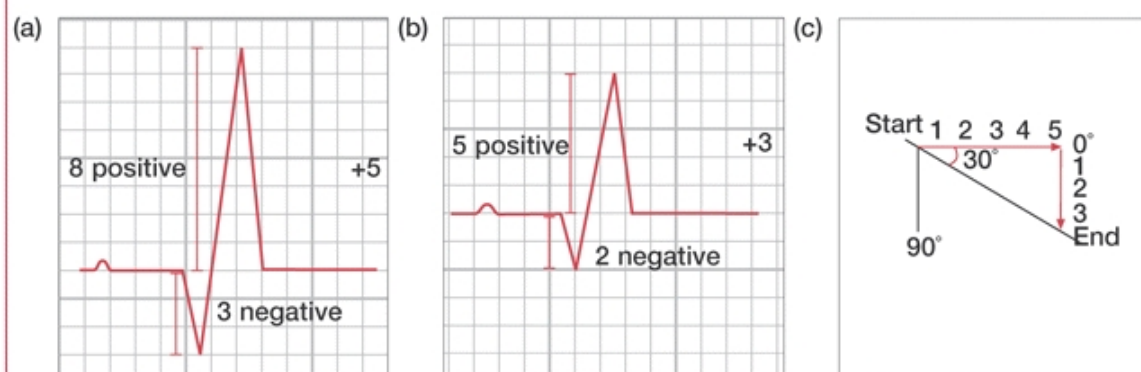


Figure 9

Vector diagram to determine the QRS axis.

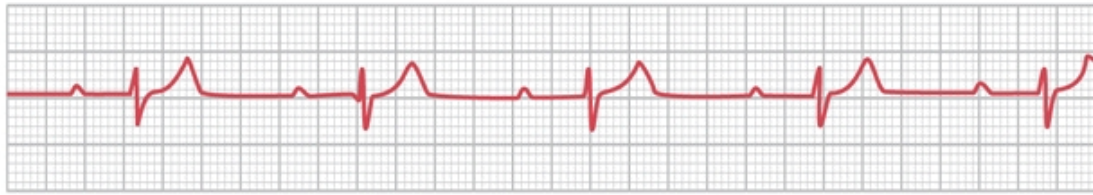


Figure 10

ECG demonstrating first-degree heart block.

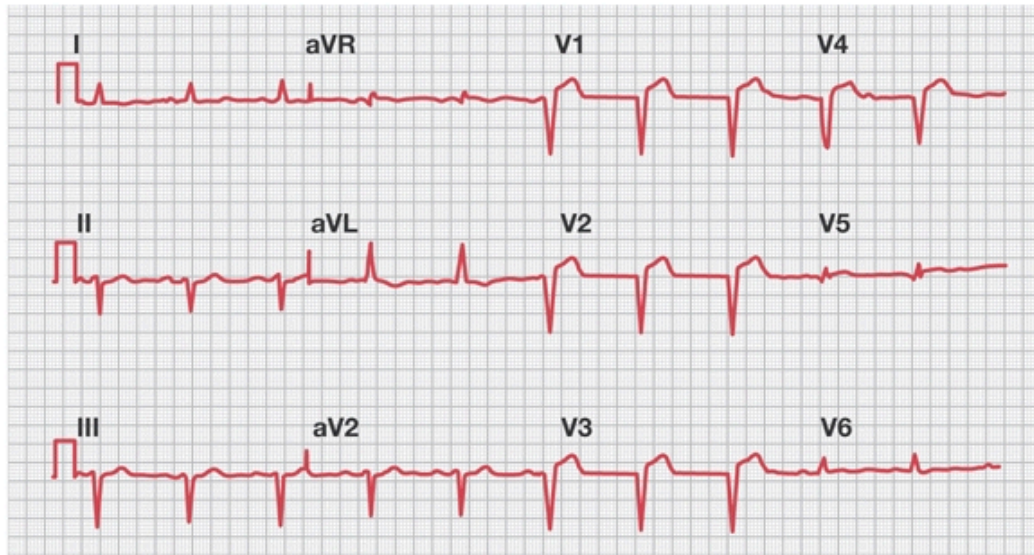


Figure 11

ECG demonstrating abnormal Q waves in V1–V4. This is indicative of a previous infarction.

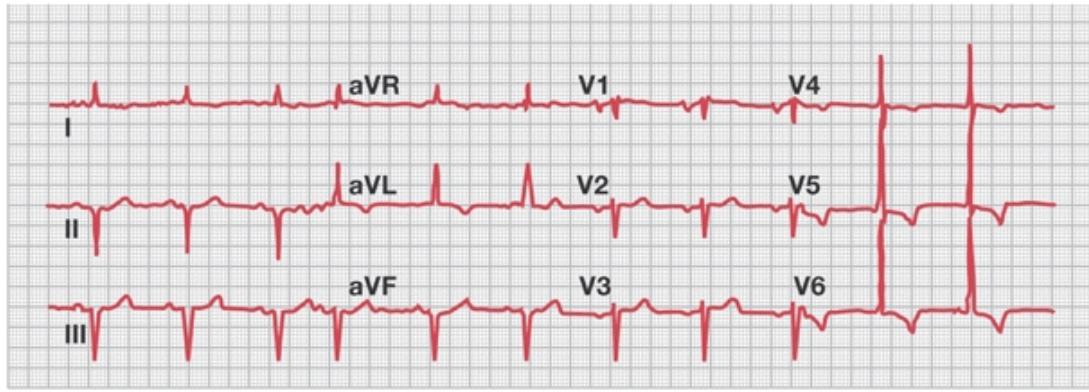


Figure 12

ECG demonstrating left ventricular hypertrophy. Note also the T-wave inversion in leads V4–V6. This is often labeled "strain".



Figure 13

ECG demonstrating left bundle branch block.

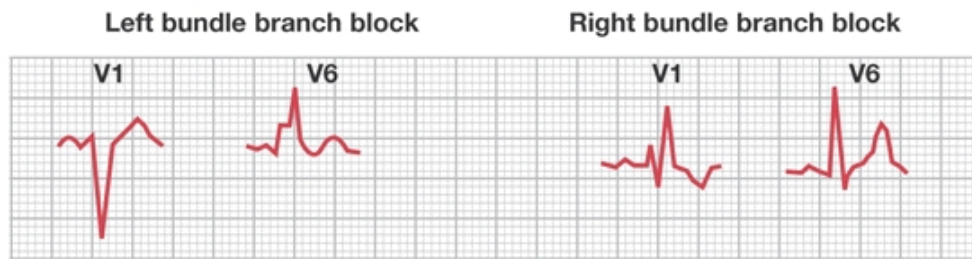


Figure 14

The shapes of V1 and V6 QRS complexes in left and right bundle branch block.

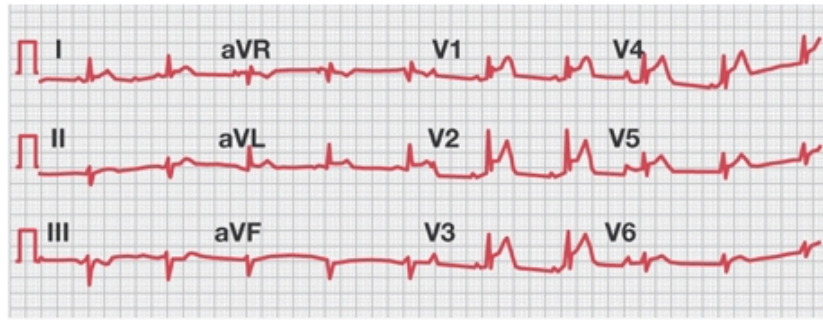


Figure 15

ECG demonstrating anteroseptal myocardial infarction. Note the ST-segment elevation.

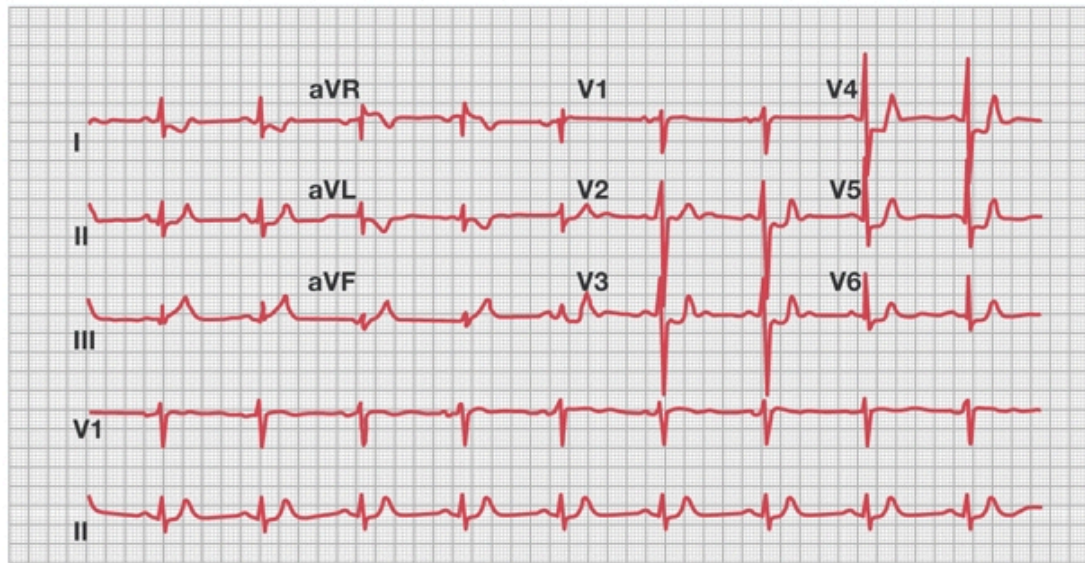


Figure 16

ECG demonstrating ST-segment depression (I, V3–V6).

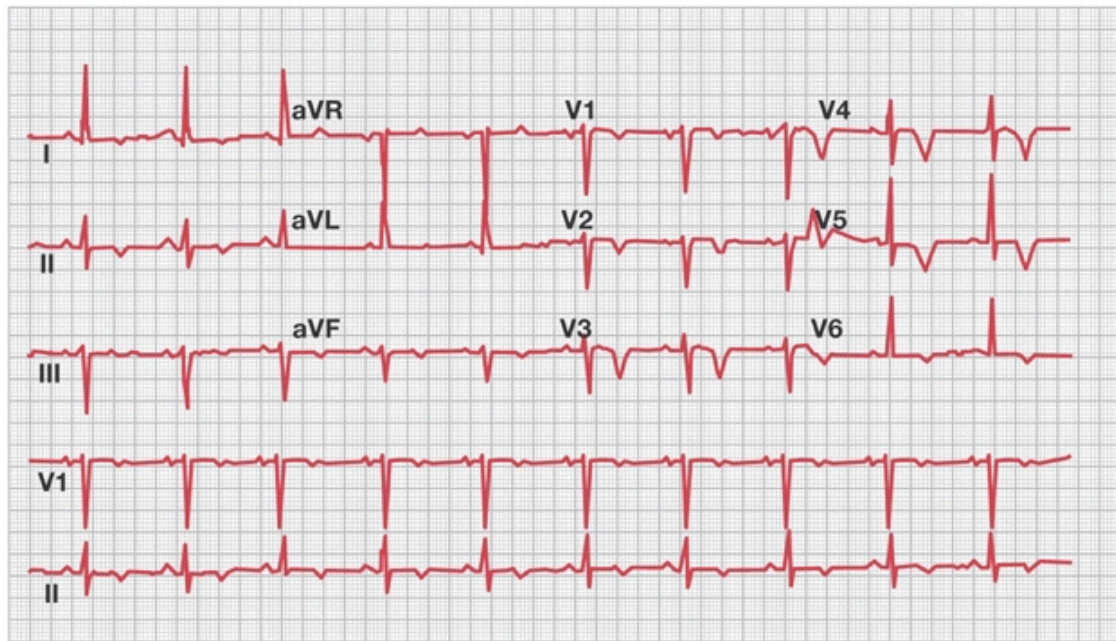


Figure 17

ECG demonstrating T-wave inversion.

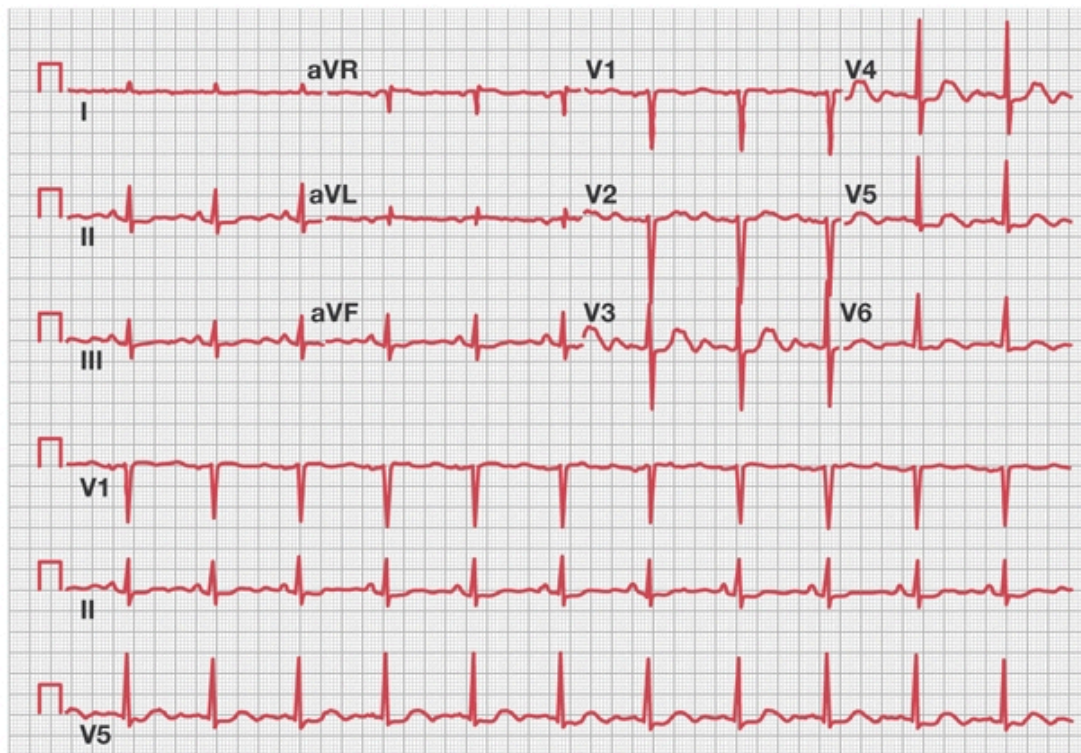


Figure 18

ECG demonstrating a long QT interval.

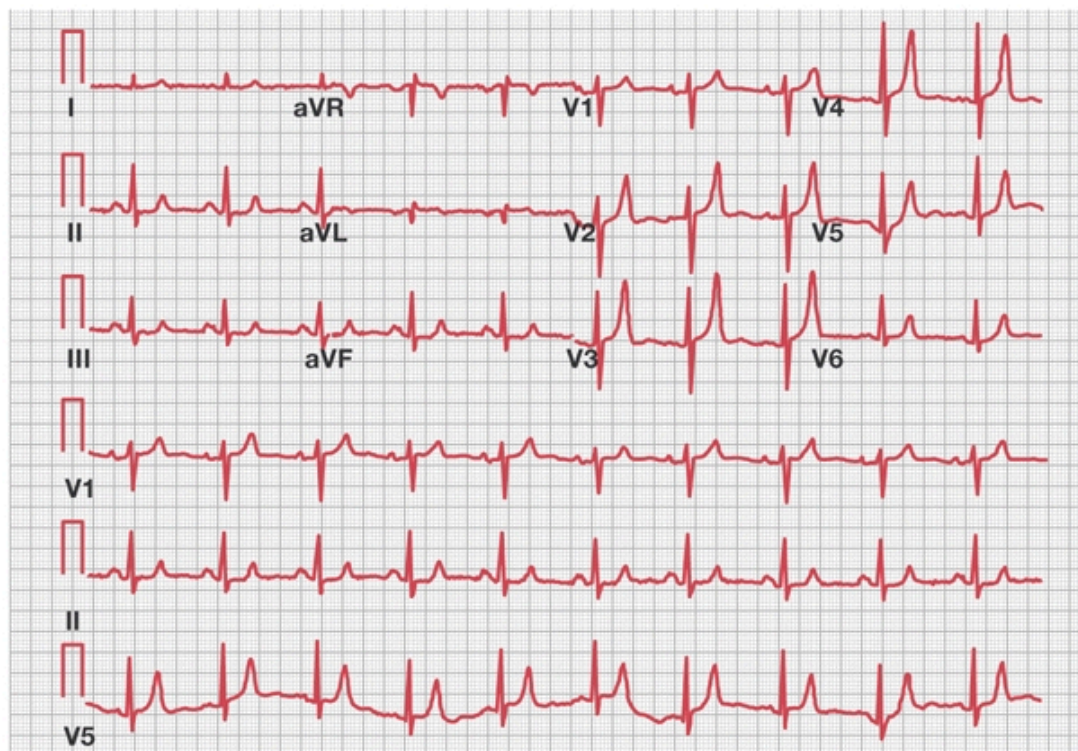


Figure 19

Hyperkalemia. Note the tall, peaked T waves.

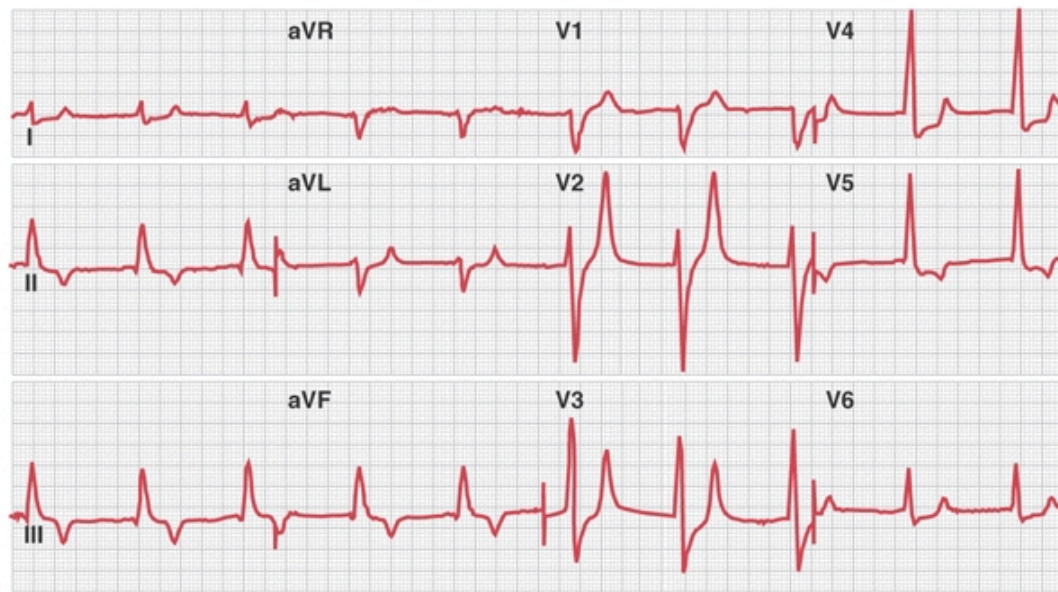


Figure 20

ECG demonstrating a widening of the QRS complex.

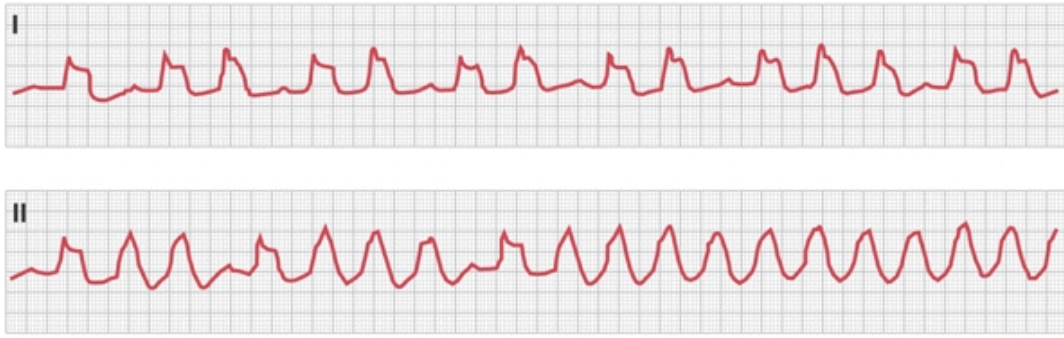


Figure 21

ECG demonstrating a sinus-wave QRS pattern.

Tables

Table 1 ECG leads and their respective views of the heart

View	Lead
Inferior	II, III, aVF
Anterior	I, aVL, V1–V3
Septal	V3, V4
Lateral	V4–V6

Table 2 Some common heart rates as determined by analysis of the QRS complex

Number of large squares between QRS complexes	Heart rate (bpm)
5	60
4	75
3	100
2	150

Table 3 Causes of a long QT interval

Congenital	Acquired
Jervell and Lange–Nielsen syndrome	Amiodarone, sotalol
Romano–Ward syndrome	Flecainide
	Hypocalcemia
	Hypokalemia
	Hypomagnesemia
	Phenothiazines
	Tricyclic antidepressants

Table 4 Drug-induced increase in the QT interval and torsade de pointes

Generic name	QT interval	Torsade de pointes	Generic name	QT interval	Torsade de pointes
Antiarrhythmics			Selective serotonin re-uptake inhibitors		
Ajmaline	+	+			
			Fluoxetine	+	+
Amiodarone	+	+			
			Paroxetine	+	
Chinidine	+	+			
			Sertraline	+	+
Disopyramide	+	+			
Dofetilide	+	+	Anticonvulsants		
Ibutilide	+	+			
			Valproate	+	
Propafenone	+	+			
Sotalol	+	+	Other psychopharmaceuticals		
Antibiotics (macrolides)			Chloralhydrate	+	+
Azithromycin		+	Levomethadone	+	+
Clarithromycin	+	+			
			Lithium	+	
Clindamycin		+			
			Naratriptan	+	
Erythromycin	+	+			
			Sumatriptan	+	
Roxithromycin	+				
Spiramycin	+	+	Venlafaxine	+	
Antibiotics (quinolones)			Zolmitriptan	+	
Gatifloxacin	+	+	Anti-Parkinson's		
Grepafloxacin ^a	+	+			
			Amantadine		+
Levofloxacin		+			
			Budipine ^c	+	+
Moxifloxacin	+	+			
Sparfloxacin	+	+	Antimalarials		
Other antibiotics					
			Quinine	+	+
Amoxicillin	+				
			Chloroquine	+	+

Generic name	QT interval	Torsade de pointes	Generic name	QT interval	Torsade de pointes
			Halofantrine	+	+
Trimethoprim-sulfamethoxazole	+	+			
			Mefloquine	+	
Antihistamines					
Astemizole ^a	+	+	Diuretics		
Clemastine	+		Indapamide	+	
Diphenhydramine	+				
Hydroxyzine	+		Lipid-lowering agents		
Terfenadine	+	+	Probucol	+	+
Antidepressants					
			Motility enhancers		
Amitriptyline	+	+			
			Cisapride ^a	+	+
Clomipramine	+				
Desipramine	+	+	Nootropic geriatrics		
Doxepine	+		Vincamine	+	+
Imipramine	+	+			
			Chemotherapeutics		
Maprotiline	+	+			
			Tamoxifen	+	+
Neuroleptics					
			Pentamidine	+	+
Amisulpride	+				
Clozapine	+		Immunosuppressants		
Chlorpromazine	+	+			
			Tacrolimus	+	+
Droperidol ^a	+	+			
Fluphenazine	+		Peptides		
Haloperidol	+	+			
			Octreotide	+	
Melperone	+	+			
Olanzapine	+		Virostatics		
Pimozide	+	+	Foscarnet	+	
Quetiapine	+				
			Muscle relaxants		
Sulpiride	+	+			
Thioridazine	+	+	Tizanidine	+	

Generic name	QT interval	Torsade de pointes	Generic name	QT interval	Torsade de pointes
Risperidone	+				
			X-ray contrast agents		
Sertindole ^b	+	+			
Tiapride	+	+	Ioxaglate meglumine	+	+
Trazodone	+				

+ A prolonged QT interval can occur or torsade de pointes was observed

a Taken off the market.

b Suspended from the market, final decision by the regulatory authorities still awaited.

c Indication limitations have been expressed.

Important tips on the use of the table: information is based on the latest scientific knowledge as far as it is generally available from published studies (Medline research), case reports, internet publications, specialist information, the Red List, and information from the regulatory authorities. In the case reports available about torsade de pointes, the causal relationship to the ingestion of the particular medication is no longer apparent; pure coincidence cannot be excluded in individual cases.

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